

University of Wah

Department of Computer Science



Submit to:

Ms. Wasifa Kanwal

Submit by:

Muhammad Talha

Course:

Information and Security Lab - CS-251L

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

1. Confidentiality:

Meaning:

Confidentiality means keeping information private and secure so that only authorized people can access it. The main goal is to protect sensitive information from being shared with the wrong person.

Real-life Example:

When you enter your ATM PIN at a cash machine, only the bank system and you should know it. If someone else sees or gets your PIN, confidentiality is broken.

2. Integrity:

Meaning:

Integrity means keeping the information accurate and unchanged. It ensures that data is not modified, deleted, or tampered with without permission.

Real-life Example:

If your bank balance is Rs10,000 and someone tries to change it to Rs1,000 without your permission, that would be a violation of integrity. The correct balance must stay the same unless you make a transaction.

3. Availability:

Meaning:

Availability means making sure information and systems are accessible whenever needed by authorized people.

Real-life Example:

When you open a bank's mobile app to check your account, it should work 24/7. If the system goes down and you can't access your money, availability has failed.

Task 02:

List five vulnerabilities that can exist in a computer lab?

Answer:

1. Unauthorized physical access:

strangers can use or steal computers, USBs, or papers.

2. Out-of-date software / missing patches:

OS or apps not updated, exploitable by malware.

3. Weak or shared passwords:

everyone uses same password or simple passwords that are easily guessed.

4. Infected removable media (USBs):

students plug infected USBs and spread malware.

5. Poor network segmentation:

all lab PCs on same network allowing one infected machine to attack others.

Task 03:

Research two famous cyber attacks and describe what happened?

Answer:

1. WannaCry (May 2017):

- **How it spread:** WannaCry was a ransomware worm that used a Windows vulnerability called EternalBlue (leaked NSA exploit) to automatically spread between machines on the internet and local networks.
- **Impact:** It infected hundreds of thousands of computers across ~150 countries and hit hospitals, companies and governments — notably the UK NHS where appointments and services were disrupted. The attack caused major operational disruption and financial cost.
- **Why it mattered:** Because it combined a wormable exploit + ransomware, unpatched systems spread it quickly. Many affected machines were running unpatched Windows versions.

2. SolarWinds / Solorigate (discovered Dec 2020):

- **How it worked:** Attackers inserted a backdoor into legitimate updates of SolarWinds' Orion IT-management software. When thousands of customers installed the trojanized update, attackers gained access to many government and private networks.
- **Impact:** High-profile US federal agencies and large private companies were compromised; the breach was stealthy and persisted for months before detection. It's treated as a major supply-chain intrusion that led to wide-reaching espionage and data access.
- **Why it mattered:** A trusted vendor update was the delivery vector meaning even secure networks were at risk if they trusted the compromised software.

Task 04:

Make a small table showing preventive and detective security controls with one example each?

Answer:

Type of Control	Purpose	Description	Example
Preventive security Control	Stop or reduce security incidents from happening	Implemented to prevent attacks or	Strong password policy / firewall

		unauthorized actions before they occur	
Detective Security Control	Identify or detect incidents after they happen	Helps to discover or alert when a security event or attack occurs	Intrusion detection system (IDS) / CCTV

Task 05:

Suggest three ways to improve security in your university labs?

Answer:

1.Strong Authentication:

- Use unique usernames and strong passwords for lab computers and Wi-Fi.
- Example: Every student has their personal login; no shared passwords.

2.Install Security Software:

- Install antivirus, anti-malware, and firewalls on lab PCs.
- Regular updates and scans help prevent malware from spreading.

3.Physical Security & Access Control:

- Restrict lab entry to authorized students and staff only.
- Use CCTV cameras and avoid letting unauthorized devices connect.

Task 06:

How can hacker use social engineering to trick someone?

Answer:

1.Phishing (Email / SMS / WhatsApp):

- *Example:* A fake email says “Reset your university password” with a link.
- *How to spot / defend:* Check sender address, hover links to see real URL, don’t enter credentials on pages reached from unexpected messages. Verify via official channels.

2.Pretexting (made-up story to build trust):

- *Example:* Someone calls pretending to be IT staff asking for login to “fix an issue.”

- *How to spot / defend:* Always verify caller identity (call back on an official number). Legitimate IT will not ask for your password.

3.Baiting (offering something to lure you):

Example: A USB found in lab labeled “Exam answers” — someone plugs it in.

- *How to spot / defend:* Never plug unknown USBs; scan removable media on isolated machines; report found devices to admin.

Task 07:

Create a simple security policy for a small business (include a password rules, software updates etc)?

Answer:

Small Business Security Policy:

1.Password Rules:

Use strong passwords: at least 8 characters, mix of letters, numbers, symbols.

Change passwords every 3 months.

Never share passwords with others.

2.ftware Updates:

Keep all operating systems, applications, and antivirus software up-to-date.

Enable automatic updates where possible.

3.Data Backup:

Backup important business data daily.

Store backups offsite or in the cloud.

4.Access Control:

Only authorized employees can access business systems and sensitive data.

Use role-based access: employees see only what they need for their job.

5.Physical Security:

Lock offices and storage areas after working hours.

Restrict access to computers when unattended (use screen locks).

6.Employee Awareness

Train staff to recognize phishing emails, suspicious links, and malware.

Report any security incidents immediately to management.